



Address:
**6745 Amberle St
San Diego, CA 92120**

Prepared for:
Ryan Feeney



Prepared by:



SUSTAINRGY

2358 University Ave., PMB 1865
San Diego, CA. 92104

www.sustainrgy.com



General



General Information

EE Impact

The climate zones are the geography grouped by seasonal weather, including temperature, humidity, and altitude. These zones are determined by weather stations serving that zone. Many of these zones are grouped by defining building requirements in the energy code. An example is attic insulation, for climate zone 7(coastal), min insulation is R19 versus climate zone 10(desert) is R30.

Location - Hotter Climate: This will result in higher energy costs during the Summer and Fall seasons.

Building Type

This property is a 1 story

Orientation

The orientation of a house is an essential consideration due to the significant contribution of sun-radiant heat gain to the dwelling. The dwelling South and West facing sides will experience the highest sun heat gain, while the East side experiences sun heat gain in the mornings. On the other hand, the North side of the dwelling never receives direct sun exposure. This can lead to a cold comfort condition.

Age

Year built: 1971

Goals/Needs

Customer Goals/Needs - High Energy Costs Comfort Issues Location(s): Living room

Renovations

None planned. New AC system, New solar system installed in 2025

All-Electric Design

Impact

Global and national agencies mandate a reduction of CO2 emissions causing the greenhouse gas effect. This underscores the necessity of an all-electric design in residential dwellings for new and existing dwellings. This all-electric design, which converts all major gas-fueled appliances to electrical ones for household heating, water heaters, clothes dryers, and kitchen cooktops, is crucial in our fight against climate change.

The electrical capacity of existing residential households has become a crucial limiting factor in conversion to an all-electric energy source for major household appliances.

This will require a higher capacity for the central electrical panel. The past minimal standards for households are 100 amps. Any new construction or conversion to high-capacity electrical will require this.

Homeowners are not alone in this transition. They are encouraged and incentivized with rebates and tax credits to switch to electric appliances, such as heating and cooling systems, water heaters, clothes dryers, and kitchen cooktops. These electrical appliances require a high voltage of 240v service with 30 amp or higher capacity, and the support is there to help make this transition smoother.

The energy assessment will include evaluating the electrical capacity at the panel and recording any open slots for breaker capacity. Also, note any high-capacity breakers installed and if they are in use.

Conditions

Electrical panel capacity - 200 amp

EV(s) noted with charger. **Breaker space** - No or limited breaker space

Gas fueled appliances , -Heating furnace, and Cooking stove

**Renewables**

Installed, Leased Year Installed: 2025



This is an assessment of this dwelling's energy efficiency. Energy efficiency impacts energy costs, comfort, health & safety issues. The assessment includes conditions and recommendations.



Building Envelope

EE CMF Hlth

EE Impact

The building envelope affects energy efficiency, comfort, and health & safety. This assessment includes an overall roof, siding, windows, foundation, insulation, attic ventilation, and indoor air infiltration. All these components are interrelated and are evaluated to provide an overall energy assessment. An example condition could be a high glass/wall ratio facing West that would result in high afternoon sun heat gain, creating comfort and high energy bills.

EXTERIOR

Orientation

FRONT facing West Room associated with this side - Kitchen
Type - Stucco over Frame



High Glass Ratio

Glass is a poor insulator that can contribute to comfort conditions and higher energy costs.

Assessment - Minimal glass/wall ratio.



Shading

Noted shading (structural, landscaping). This will effectively reduce sun heat gain. This can reduce heat buildup in walls improving comfort and reducing energy costs associated with comfort.

**Recommendation**

Recommendation(s) - No recommendations for exterior walls.

EXTERIOR**Orientation**

SIDE facing South Room associated with this side - Bedrooms

**High Glass Ratio**

Assessment - Minimal glass/wall ratio.

Shading

Minimal or no shading noted on this wall

Recommendation

Recommendation(s) - No recommendations for exterior walls.

EXTERIOR

Orientation

REAR facing East Room associated with this side - Living, family Bedrooms

Type - Stucco over Frame

**High Glass Ratio**

Glass is a poor insulator that can contribute to comfort conditions and higher energy costs. **Assessment** -

Noted a moderate glass/wall ratio on this side of the dwelling.

Glass is a poor insulator that will contribute to comfort conditions and higher energy costs.

**Shading**

Noted shading (structural, landscaping). This will effectively reduce sun heat gain. This can reduce heat buildup in walls improving comfort and reducing energy costs associated with comfort.

**Recommendation**

Recommendation(s) - Recommend structural or landscape shading to reduce the sun heat gain. This can be awnings, patio covers or landscaping. Deciduous trees can be advantageous as they shed leaves in the Fall that allows sun heat gain during the Winter and Spring seasons.

EXTERIOR

Orientation

SIDE facing North

**High Glass Ratio**

Glass is a poor insulator that can contribute to comfort conditions and higher energy costs. **Assessment** - Minimal glass/wall ratio.

Shading

Minimal or no shading noted on this wall

Recommendation

No recommendations for exterior walls.

**Windows****EE Impact**

Windows offers many benefits of ventilation, natural light, scenic views, and architectural aesthetics. But the downside is that windows are poor insulators, leading to a lack of energy efficiency and comfort issues. Windows with a high glass-to-wall ratio will further impair these conditions. Even dual-pane windows have poor efficiency compared to an insulated wall. South and West facing walls with moderate glass exposure can create an overheated comfort condition. Conversely, a North-facing wall with a high glass ratio can create a cold comfort condition. Lastly, older single-pane windows have poor energy efficiency and lack air tightness creating draft conditions.

Type

Noted all dual-pane windows

**Assessment**

Good - Noted dual-pane windows. Recommend to consider tinting, shading and blinds to reduce sun radiation heat gain to improve efficiency and comfort.

Recommendation

No recommendation for this condition.

**Insulation****EE Impact**

Insulation is a fundamental component of a well-constructed dwelling that provides thorough thermal protection with numerous benefits, including energy efficiency, comfort, moisture barrier, and sound control. Insulation must be fully fluffed, continuous with no gaps providing an effective thermal and air barrier to the conditioned living space. Proper installation and maintenance of insulation are essential to ensure it functions effectively. and provides all

these benefits.

Attic

Multiple roof/attic systems. Mix of standard and cathedral/vaulted ceilings. Portions of ceilings are not accessible.
Install Quality: Poor

Assessment

Attic - Original installation. Most of the insulation show degradation and compression. Missing sections on knee wall and displacement.

Install Quality: Poor Recommend to remove existing insulation, air seal and install new insulation to R30 levels.



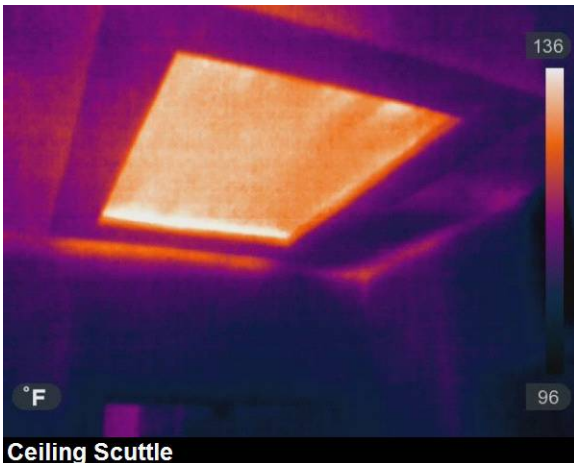
Knee walls missing insulation



Missing insulation

Recommendation

Attic cover should be insulated. Moderate thermal break of heating and cold air penetrating conditioned area. **Recommend** correction



Ceiling Scuttle

Wall - Exterior

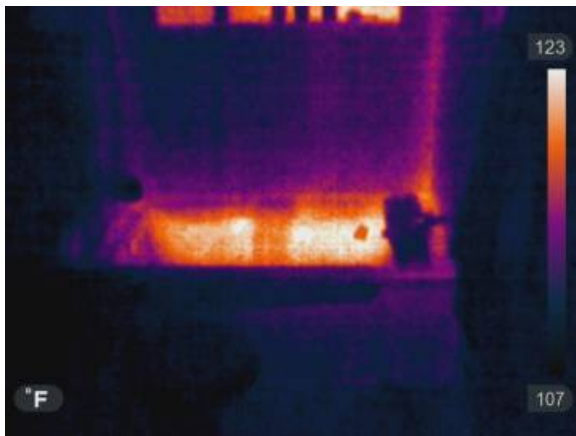
Wall insulation plays a major role in energy efficiency. The following is general information based on the time of construction. These conditions can change in repairs or remodeling.

Type

Insulation type - No insulation noted. Proper installation of insulation to increase the energy efficiency of the house is recommended.

Assessment

Thermal scan revealed several conditions indicating lack of insulation in walls and/or ceiling. This will impede thermal protection that will contribute to comfort issues.

**Recommendation**

Recommend complete installation of blown-in insulation into all exterior walls. This will impede outside temperatures improving energy efficiency and enhancing comfort. This will also act as a sound barrier from outside noise.

**Foundation**

Slab. No impact on energy efficiency.



Ventilation

EE Impact

Attic ventilation mainly facilitates airflow to manage heat buildup. The benefits include the reduction of heat transfer to both the living area and the roof system.

Type

Lower vents - Noted lower vents Upper vents - Noted upper vents and appears to be operating properly



Lower vents - Noted lower vents Upper vents - Noted upper vents and appears to be operating properly



Assessments

OK. There is sufficient ventilation for attic/roof system.

Recommendations

OK. There is sufficient ventilation for attic/roof system.



Comfort Recommendation - Recommend a whole-house fan system. These systems are ideal by providing cooling comfort for the living areas of the dwelling when cooler outdoor temperatures exist. This is highly energy efficient compared to AC and more effective than ceiling fans.



Whole-house Fan - <https://quietcoolsystems.com/>



Air Sealing

EE Impact

The purpose of a ventilation system is to ensure a controlled exchange of indoor and outdoor air. Airtight homes, designed to minimize the infiltration of outdoor air, can suffer from insufficient fresh air and the accumulation of indoor pollutants, moisture, and odors. A well-designed ventilation system addresses these issues and serves several important functions: habitability, Indoor Air Quality (IAQ), moisture control, odor removal, and energy efficiency.

1. **Health and Comfort:** Proper ventilation provides fresh air, essential for comfort and well-being, reducing health risks from poor indoor air quality.
2. **Indoor Air Quality (IAQ):** Ventilation removes these, maintaining a healthy environment.
3. **Moisture Control:** Ventilation manages humidity, expelling excess moisture from bathrooms and laundry rooms.
4. **Odor Removal:** Ventilation helps eliminate odors from cooking and pets, enhancing home comfort.
5. **Energy Efficiency:** A properly managed ventilation system optimizes energy efficiency within the mechanical system, ensuring comfort, regulating airflow, and more.

Exterior Doors

Front Door Noted weather-stripping appears serviceable. Gaps between door and frame allowing air leakage - None



Door to Garage - Noted weather-stripping installed Gaps between door and frame allowing air leakage - None



Attic scuttle

**Assessment**

OK - Exterior doors properly weather-stripped and air tight.

**Recommendation(s)**

No recommendation for this condition.

IAQ/CAZ

Impact

Numerous statistics highlight the importance of indoor air quality within a home. Factors such as mold, off-gassing from new materials including furniture and flooring, and moisture management play crucial roles. This assessment is limited on identifying exposure to carbon monoxide, acknowledging that past building practices may contribute to this risk.

IAQ - Combustion

Indoor Air Quality inside a home can expose homeowners to unhealthy and hazardous conditions. This assessment is limited to carbon monoxide(CO) exposure. Past building practices can lead to exposure of hazardous carbon monoxide(CO) from combustible appliances inside the living area of a home. CO gases can be emitted from any of the following: gas furnace, gas water heater, gas stove, gas fueled fireplace and direct access to the garage. The following are conditions that warrant further evaluation: Cooking Ventilation - Properly vented to the outside.



Assessment

This dwelling has minimal exposure to CO and other unhealthy/hazardous conditions inside the living condition areas

Combustion Appliances

Heating furnace Type Central located in a utility cabinet inside the living area source combustion air from the outside. Water Heaters Electric resistance strips or heat pump

Appliances

Kitchen cooktop Natural Gas/Propane Kitchen cooktop ventilation Ducted Clothes dryer Natural Gas/Propane Installed in garage

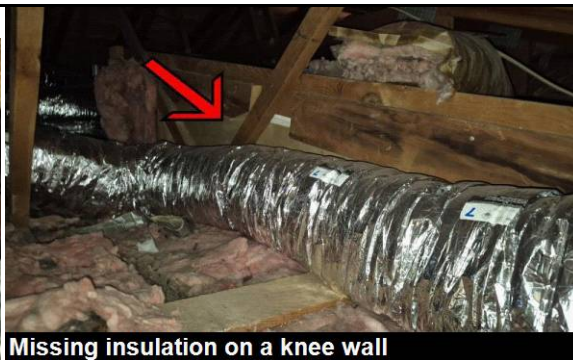
Recommendations

This dwelling has minimal exposure to CO and other unhealthy/hazardous conditions inside the living condition areas

Type



Insulation



Missing insulation on a knee wall

Blower Door Assessment

Blower door tests for home air leakage between outdoor & indoor conditioned area. Leaky (drafty) dwelling creates multiple problems. High leakage will loose conditioned air (heat/cooling) leading to higher energy costs. It also can draw in outside air that may contain outside pollutants creating an enviromental hazard.

House Blower Door Pressure Test

Test

Leakage ratio is the percentage of actual leakage compared to target. A low ratio indicates a more airtight home. A high ratio indicates high leakage that leads to hot/cold conditioning loss and draftiness. Dwelling tested at **1,211** cfm

Target is based on minimum airflow requirements determined by cubic capacity of conditioned area. Target is **2,425** cfm. Leak ratio percentage is 52% below target.



Assessment

Tight - This is below the minimum standards that can present unhealthy conditions in maintaining moisture, indoor air quality, odors, and proper access to fresh air for health and comfort.



Recommend to install an IAQ Fan system in hallway bathroom. This will intermittently draw air through the building envelope to assure sufficient air and exhaust moisture and indoor pollutants.



Mechanicals



HEATING/COOLING/DUCTING

Impact

Providing comfort to your home can be your highest utility cost. These costs vary widely based on your climate, building energy efficiency, and your HVAC systems efficiency. Over time, standards have continued to improve efficiency and effectiveness to achieve comfort cost-effectively. Newer systems have greatly improved efficiency. The new AC SEER 14 is over 57% more efficient than an end-of-life SEER 8 unit.

With the advent of Climate Change, new government standards are focused on greenhouse gas reduction. These standards now specify all electric solutions around heat pump technology in heating/ cooling systems, water heaters, and clothes dryers.

As a result, replacing older, less efficient heating/cooling systems should be considered.

Numerous incentives in tax and rebates are available. This includes residential conversions to an all-electric design, energy efficiency, sustainability, and more.



Heating

EE Impact

Providing comfort to your home can be your highest utility cost. These costs vary widely based on climate, HVAC equipment efficiency, utility costs, and building envelope.

For decades gas, gas-fueled draft-assisted central furnaces have been installed due to 80% AFUE, and natural gas has lower energy costs compared to electric. The high-efficiency 90%+ AFUE gas furnaces have high equipment costs that impede their cost-effectiveness.

Energy costs can unexpectedly rise due to natural gas and electricity market changes. Staying informed and proactive can help minimize the impact, explore alternative energy sources, and invest in energy-efficient technologies.

Investing in alternative energy sources and energy-efficient technologies is crucial to mitigate Climate Change and reduce greenhouse gas emissions. New building standards now prescribe energy-efficient electric heat pump furnaces.

Numerous incentives in tax and rebates are available. This includes residential conversions to an all-electric design, energy efficiency, sustainability, and more.



Type

Natural Gas or propane **Type** - Central - Draft Assist unit has an fan to optimize air through the heat exchange.

Brand/Model # - Day Night. Energy Efficiency rating is 80% AFUE (Annual Fuel Utilization Efficiency). Mfg: 2023



Assessment

Standard gas heating unit is in serviceable condition.

Recommendation

Good - meets energy efficiency standards

Thermostat

Noted a programmable thermostat. These should be programmed based on day/night, seasons and occupancy. Please refer to thermostat manual for instructions.



Air Conditioning

EE Impact

The performance of a cooling system can significantly impact its energy efficiency and costs. Climate conditions and building exterior efficiency also play a crucial role in determining the system's overall performance and energy consumption. Manufacturers have made significant strides in improving cooling system efficiency in recent years. The current SEER 14 standards are 57% more efficient than the SEER 8 systems built in the 1970s.

Type - Condenser



Type**Type** - Condenser

Capacity (tons) - 4 tons

SEER Rating - +14

Assessment

SEER rating is estimated to be 14 or higher based on label or age of unit. Good. Unit is newer and energy efficient.

**Recommendation**

Good - meets energy efficiency standards

**Ducting****EE Impact**

Building standards for decades had no minimums for ducting air tightness. The Energy Commission has assessed older ducting designs leak an average of 30%. This inefficiency contributes to wasted energy costs and depressurizes the living areas, contributing to unhealthy and hazardous conditions.

Type

Duct Type - Noted newer flex ducting. New flex ducting systems are highly efficient when installed properly. These systems have higher insulation layers and should leak 5% or less.

**Assessment - Supply**

Highly insulated, airtight ducting. Energy efficient

**Recommendation Supply**

Recommend to seal registers to wall/ceiling to reduce duct leakage and improve efficiency.

**Assessment - Return**

Return - Ducted

Recommendations

Unit is professionally sealed and airtight. This improves energy efficiency to your HVAC system.



Water Heater

Impact

Water heaters rank second behind heating and cooling systems in terms of energy costs. This highlights the importance of optimizing their efficiency to reduce energy costs, water conservation, and environmental impact. Water heaters have a finite lifespan, and as they age, they become less efficient. Replacing them every 10-15 years is essential to maintain energy efficiency and prevent potential sudden breakdowns that result in higher replacement costs for emergency services.

Lo-Flo water fixtures and insulation of hot water lines should be considered to optimize energy efficiency and water conservation. Recirculation pumps (wireless activated) should also be considered for long runs or to experience long delays for hot water service.

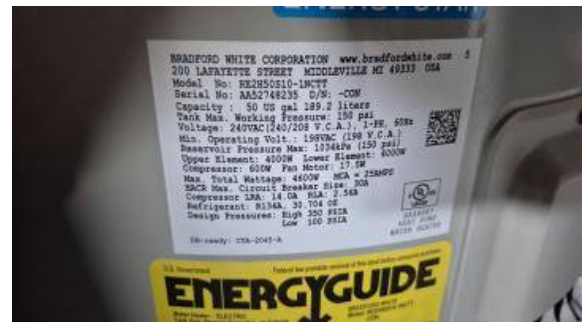
Electrical Heat-Pump Water Heaters (HPWH) have become the building standard and should be considered for replacement. The savings are significant. On average, a family of four can save \$330 a year in energy costs. That is \$3,530 over its lifetime. As these heat pump water heaters have higher upfront costs, their payback is within 2 years.

Type

Fuel - Electric **Location** - Adjacent - Unit is located in a adjacent area like garage to the conditioned area. There is a potential of back-drafting of combustion gases and carbon monoxide. This is dependent of high levels of depressurization within the conditioned area. **Type** - Brand/Model # - Hpwh



Type - Brand/Model # - Bradford White



Assessment

Appears serviceable. Meets or exceeds current energy standards.





Appliances/Lighting/Plug-ins



Appliances

EE Impact

Kitchen and laundry appliances are crucial to a household's overall energy efficiency. By choosing energy-efficient appliances, performing regular maintenance, and adopting energy-saving habits, consumers can significantly reduce their energy consumption and contribute to environmental sustainability.

Refrigerators stand out for high energy costs. This big consumer has been compounded by the additional need for more storage (garage) and compact refrigeration appliances for compact wine and beverages. Highly energy-efficient models should be considered, saving hundreds of dollars annually and a short payback.

In an effort to reduce the greenhouse effect, it is recommended to consider switching to highly energy-efficient electric appliances or gas cooking stoves and clothes dryers. Electric induction stove and electric heat pump clothes dryer stand out for their impressive energy efficiency.

Energy Star standards are established by EPA and Dept of Energy (DOE) to promote energy efficiency. The Energy Star label promotes environmentally and saves money on energy bills. These standards must meet strict standards requiring testing by EPA, which is more stringent than national standards. It also promotes performance and durability.

Kitchen

Refrigerators - Appears to be a newer model. These are moderately more efficient compared to models 10 years or older. Noted multiple refrigerators/freezers. If older than 10 years old, recommend to upgrade to energy efficient models.





**Clothes W/D
Clothes Washer -**



Assessment

Refrigerator - Refrigerator - Noted older secondary refrigerator that lacks energy efficiency compared to newer models.



Lighting

EE Impact

Fixed lighting consumes an average of 11% of overall energy usage. LED has dominated the lighting fixtures market due to significant energy savings of 75%, durability lasting 25x longer, and rich in features. Light switch technology is part of the energy efficiency for lighting. Dimmable switches are required for habitable rooms and hallways. Indoor utility rooms must now utilize occupancy sensor switches, including bathrooms, laundry, and garages. Outdoor lights should have a photo and occupancy sensor.

Type

Lighting Noted mostly LED hi-efficacy lighting. Noted dimmable switches that will reduce energy to fixed lighting.



Noted dimmable switches that will reduce energy to fixed lighting.



Noted outdoor photo sensors to reduce energy consumption.



Recommendation

Indoor Sensors - Recommend to install indoor occupancy switch sensors in bathrooms and utility room to reduce energy consumption.



Other

Impact

Plug-ins are devices and appliances that draw their energy from an electrical outlet. The rapid adoption of technology has resulted in a significant increase in energy consumption. New appliance standards enforce energy efficiency features.

The energy consumption of plug-ins comes down to the homeowners involvement in understanding, monitoring, and management. The following are recommendations improve energy savings with plug-ins:

- Any appliance with an Energy Star rating should be evaluated. Many times, the higher costs have a payback over a period of time, making it a smarter, lower-cost purchase.
- Look for power and energy-saving features on new purchases for computers, printers, entertainment systems, TVs, and more. Set these settings accordingly.
- Reduce standby (vampire) energy:
 - o Install smart electrical power strips.
 - o Unplug any device or appliance that has low or intermittent use.

Smart kitchen and laundry appliances can save energy with monitoring features and sensors. They can also save energy to minimize operation during peak utility rates.

Energy Monitoring Units

There are various tools available that can help in monitoring electrical energy usage in real time. Energy monitoring units (EMUs) display real-time energy usage through numeric or graphic displays. These EMUs can be installed either at the electrical panel or wirelessly via a smart outlet device at electrical outlets. This allows you to gain specific insight into your energy usage and manage it effectively by tracking baseline usage and identifying energy-consuming devices.

Assessment

Minimal



Solar/Battery

Impact

Renewable energy will continue to be a growing and dominant residential energy source. The demand is driven by cost savings, sustainability interests, all-electric building standards, and the adoption of EVs in the automotive industry. The rising demand for battery systems is driven by cost-effectiveness through time-of-day load shifting and an emergency backup source.

There are ambitious global, national, and state mandates for adopting renewable energy. For example, California has a goal of 100% renewable energy by 2045. These are large-scale significant challenges, including technological,

economic, and regulatory hurdles.

Numerous rebates and tax incentives are available through national and state programs

Photo Voltaic

Noted photo voltaic system.



Assessment

Installed

Recommendation

Photo Voltaic system should be cleaned periodically to optimize performance. **Recommend** to have panels professionally cleaned every two years. Recommend to monitor system for performance and maintenance. This can be accomplished through energy management systems.



Battery Storage

Battery storage offers both emergency backup and load balancing of energy usage. Load balancing allows to utilize energy from battery storage during peak demand and rates that will reduce costs and transmission back to the utility provider.



Summary of Recommendations



The following are recommendations to improve: energy efficiency, comfort, health & safety conditions.

Recommendations

BUILDING

INSULATION

Assessment

Attic - Original installation. Most of the insulation show degradation and compression. Missing sections on knee wall and displacement.

Install Quality: Poor Recommend to remove existing insulation, air seal and install new insulation to R30 levels.

Recommendation

Attic cover should be insulated. Moderate thermal break of heating and cold air penetrating conditioned area. **Recommend** correction

Recommend complete installation of blown-in insulation into all exterior walls. This will impede outside temperatures improving energy efficiency and enhancing comfort. This will also acts as a sound barrier from outside noise.

BUILDING VENTILATION/IAQ

Recommendations

Comfort Recommendation - Recommend a whole-house fan system. These systems are ideal by providing cooling comfort for the living areas of the dwelling when cooler outdoor temperatures exist. This is highly energy efficient compared to AC and more effective than ceiling fans.

Whole-house Fan - <https://quietcoolsystems.com/>

Blower Door Assessment

House Blower Door Pressure Test

Assessment

Tight - This is below the minimum standards that can present unhealthy conditions in maintaining moisture, indoor air quality, odors, and proper access to fresh air for health and comfort.

Recommend to install an IAQ Fan system in hallway bathroom. This will intermittently draw air through the building envelope to assure sufficient air and exhaust moisture and indoor pollutants